Instructions:

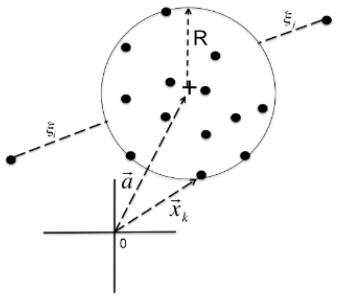
You will work on these exercises during the practical sessions in the fifth week of the course.

Below is an indication of the exercises that we will cover (at least) in each session. We encourage you try them before the session.

- First session (Wednesday/Friday): exercises 1 and 3.
- Second session (Friday/Monday): Multiple choice and True/False questions from last year's exams (available on Canvas).

Exercise 1: Kernel Outlier Detection

Consider the picture below. The dots represent data items. Our task is to derive an algorithm that detects the outliers. To that end, we draw a circle centered at location $\boldsymbol{a}$ with radius R. All data cases that fall outside the circle are detected as outliers.



We write down the primal program that finds such a circle:

$$\min_{\boldsymbol{a},R,\boldsymbol{\xi}} R^2 + C\sum_{i=1}^{N}\xi_i$$

subject to

$$\|\boldsymbol{x}_i - \boldsymbol{a}\|^2 \leq R^2 + \xi_i, \qquad \xi_i \geq 0, \qquad i = 1,\ldots,N.$$

In words, we want to minimize the radius of the circle subject to the constraint that most data cases should lie inside it. Outliers are allowed to stay outside, but they pay a price proportional to their distance from the circle boundary and C.

(a) Introduce Lagrange multipliers for the constraints and write down the primal Lagrangian. Use $\{\alpha_i\}$ for the first constraint and $\{\mu_i\}$ for the non-negativity constraint $\xi_i \geq 0$.

Solution:

Writing the first constraint as $R^2 + \xi_i - \|\boldsymbol{x}_i - \boldsymbol{a}\|^2 \geq 0$, the Lagrangian is

$$\mathcal{L}(\boldsymbol{a}, R, \boldsymbol{\xi}, \boldsymbol{\alpha}, \boldsymbol{\mu}) = R^2 + C\sum_{i=1}^{N}\xi_i - \sum_{i=1}^{N}\alpha_i\left(R^2 + \xi_i - \|\boldsymbol{x}_i - \boldsymbol{a}\|^2\right) - \sum_{i=1}^{N}\mu_i\xi_i.$$

(b) Write down all stationary and KKT conditions. Hint: take the derivative with respect to R^2 instead of R.

Solution:

The stationary conditions are obtained by differentiating with respect to R^2, $\boldsymbol{a}$, and ξ_i:

$$\frac{\partial \mathcal{L}}{\partial R^2} = 1 - \sum_{i=1}^{N} \alpha_i = 0, \qquad \text{so} \qquad \sum_{i=1}^{N} \alpha_i = 1.$$

$$\frac{\partial \mathcal{L}}{\partial \boldsymbol{a}} = -2 \sum_{i=1}^{N} \alpha_i (\boldsymbol{x}_i - \boldsymbol{a}) = 0, \qquad \text{so} \qquad \boldsymbol{a} = \sum_{i=1}^{N} \alpha_i \boldsymbol{x}_i,$$

where we used $\sum_i \alpha_i = 1$. Finally,

$$\frac{\partial \mathcal{L}}{\partial \xi_i} = C - \alpha_i - \mu_i = 0, \qquad i = 1, \ldots, N.$$

The KKT conditions are:

- $R^2 + \xi_i - \|\boldsymbol{x}_i - \boldsymbol{a}\|^2 \geq 0$ for all i.
- $\xi_i \geq 0$ for all i.
- $\alpha_i \geq 0$ and $\mu_i \geq 0$ for all i.
- $\alpha_i \left(R^2 + \xi_i - \|\boldsymbol{x}_i - \boldsymbol{a}\|^2\right) = 0$ for all i.
- $\mu_i \xi_i = 0$ for all i.

(c) Use these conditions to derive which data cases $\boldsymbol{x}_i$ will have $\alpha_i > 0$ and which ones will have $\mu_i > 0$.

Solution:

The complementary slackness conditions give three cases.
If $\boldsymbol{x}_i$ is strictly inside the ball, then $\xi_i = 0$ and $R^2 + \xi_i - \|\boldsymbol{x}_i - \boldsymbol{a}\|^2 > 0$. Hence $\alpha_i = 0$, and from $C - \alpha_i - \mu_i = 0$ we get $\mu_i = C > 0$.
If $\boldsymbol{x}_i$ is outside the ball, then $\xi_i > 0$, so $\mu_i = 0$. Hence $\alpha_i = C > 0$. The first constraint is active:

$$R^2 + \xi_i - \|\boldsymbol{x}_i - \boldsymbol{a}\|^2 = 0.$$

If $\boldsymbol{x}_i$ lies on the ball boundary, then $\xi_i = 0$ and

$$R^2 - \|\boldsymbol{x}_i - \boldsymbol{a}\|^2 = 0.$$

In this case $\alpha_i \geq 0$, $\mu_i \geq 0$, and $\alpha_i + \mu_i = C$.

(d) Derive the dual Lagrangian and specify the dual optimization problem. Kernelize the problem, that is, write the dual program only in terms of kernel entries and Lagrange multipliers.

Solution:

Using the stationary conditions $\sum_i \alpha_i = 1$, $C - \alpha_i - \mu_i = 0$, and $\boldsymbol{a} = \sum_i \alpha_i \boldsymbol{x}_i$, the

Lagrangian becomes

$$\begin{aligned}\mathcal{L} &= \sum_{i=1}^{N} \alpha_i \boldsymbol{x}_i^\top \boldsymbol{x}_i - \boldsymbol{a}^\top \boldsymbol{a} \\ &= \sum_{i=1}^{N} \alpha_i \|\boldsymbol{x}_i\|^2 - \sum_{i=1}^{N}\sum_{j=1}^{N} \alpha_i \alpha_j \boldsymbol{x}_i^\top \boldsymbol{x}_j.\end{aligned}$$

Replacing scalar products by kernel evaluations gives

$$\sum_{i=1}^{N} \alpha_i K(\boldsymbol{x}_i, \boldsymbol{x}_i) - \sum_{i=1}^{N}\sum_{j=1}^{N} \alpha_i \alpha_j K(\boldsymbol{x}_i, \boldsymbol{x}_j).$$

Therefore the dual optimization problem is

$$\max_{\boldsymbol{\alpha}} \left[\sum_{i=1}^{N} \alpha_i K(\boldsymbol{x}_i, \boldsymbol{x}_i) - \sum_{i=1}^{N}\sum_{j=1}^{N} \alpha_i \alpha_j K(\boldsymbol{x}_i, \boldsymbol{x}_j) \right]$$

subject to

$$\sum_{i=1}^{N} \alpha_i = 1, \qquad 0 \le \alpha_i \le C, \qquad i = 1, \ldots, N.$$

(e) The dual program will return optimal values for $\{\alpha_i\}$. Assume that at least one of these satisfies $0 < \alpha_i < C$. In terms of the optimal values for α_i, compute the optimal values for the other dual variables $\{\mu_i\}$. Then solve the primal variables $\{\boldsymbol{a}, R, \boldsymbol{\xi}\}$ in terms of the dual variables.

Solution:

From the stationary condition $C - \alpha_i - \mu_i = 0$,

$$\mu_i^* = C - \alpha_i^*.$$

The center of the ball is

$$\boldsymbol{a}^* = \sum_{i=1}^{N} \alpha_i^* \boldsymbol{x}_i.$$

If $0 < \alpha_i^* < C$, then $\boldsymbol{x}_i$ lies on the boundary of the ball, so

$$R^{*2} = \|\boldsymbol{x}_i - \boldsymbol{a}^*\|^2 \qquad \text{for any } i \text{ with } 0 < \alpha_i^* < C.$$

The slack variables are

$$\xi_i^* = \begin{cases} \|\boldsymbol{x}_i - \boldsymbol{a}^*\|^2 - R^{*2}, & \text{if } \alpha_i^* = C, \\ 0, & \text{if } 0 \le \alpha_i^* < C. \end{cases}$$

(f) Assume we have solved the dual program. We now want to apply it to new test cases. Describe a test in the dual space, in terms of kernels and Lagrange multipliers, that could be used to detect outliers. Students who got stuck along the way may describe the test in primal space.

Solution:

A new test case $\boldsymbol{x}_t$ is an outlier if it lies outside the learned ball:

$$\|\boldsymbol{x}_t - \boldsymbol{a}^*\|^2 > R^{*2}.$$

Using $\boldsymbol{a}^* = \sum_i \alpha_i^* \boldsymbol{x}_i$, the left-hand side can be written in kernel form as

$$K(\boldsymbol{x}_t, \boldsymbol{x}_t) - 2\sum_{i=1}^{N} \alpha_i^* K(\boldsymbol{x}_i, \boldsymbol{x}_t) + \sum_{i=1}^{N}\sum_{j=1}^{N} \alpha_i^* \alpha_j^* K(\boldsymbol{x}_i, \boldsymbol{x}_j).$$

Therefore $\boldsymbol{x}_t$ is detected as an outlier when

$$K(\boldsymbol{x}_t, \boldsymbol{x}_t) - 2\sum_{i=1}^{N} \alpha_i^* K(\boldsymbol{x}_i, \boldsymbol{x}_t) + \sum_{i=1}^{N}\sum_{j=1}^{N} \alpha_i^* \alpha_j^* K(\boldsymbol{x}_i, \boldsymbol{x}_j) > R^{*2}.$$

(g) What kind of solution do you expect if we use $C = 0$? What solution do you expect if we use $C = \infty$?

Solution:

When $C \to 0$, slack variables are essentially not penalized, so the optimizer can shrink the radius towards $R = 0$ and allow many points to violate the ball constraint.
When $C \to \infty$, violations are penalized infinitely strongly, so the solution must choose a ball large enough to contain all training data cases.

Exercise 2: L2 Support Vector Machine

Exam question

Let $\{(\mathbf{x}_n, t_n)\}_{n=1}^N$ be a set of N training pairs of feature vectors $\mathbf{x}_n$ and labels $t_n \in \{-1, +1\}$. The following is the primal formulation of the L2-SVM, a variant of the standard SVM obtained by squaring the penalty for the slack variables:

$$\begin{aligned} \min_{\mathbf{w},b,\boldsymbol{\xi}} \quad & \frac{1}{2}\mathbf{w}^\top\mathbf{w} + \frac{C}{2}\sum_{n=1}^{N}\xi_n^2 \\ \text{subject to} \quad & t_n(\mathbf{w}^\top\mathbf{x}_n + b) \geq 1 - \xi_n, \qquad n = 1, \ldots, N, \\ & \xi_n \geq 0, \qquad n = 1, \ldots, N. \end{aligned}$$

Note: the first subquestion is difficult, but it is not needed to solve the rest of the subquestions.

(a) Show that removing the constraints $\xi_n \geq 0$ does not change the optimal solution to the primal problem (i.e., proof that the constraints $\xi_n \geq 0$ are no longer necessary in this L2-SVM formulation).

Hint: assume there is an optimal solution such that $\xi_n < 0$ for some j and try reach a contradiction.

Solution:

Suppose we remove the constraints $\xi_n \geq 0$ and obtain an optimal solution $(\mathbf{w}^*, b^*, \boldsymbol{\xi}^*)$. Assume for contradiction that there exists some j such that $\xi_j^* < 0$. Then, setting $\tilde{\xi}_j = 0$

(and keeping all other variables fixed) yields another feasible slack value, as

$$t_j(\mathbf{w}^{*\top}\mathbf{x}_j + b) \geq 1 - \xi_j > 1 - \tilde{\xi}_j = 1,$$

and therefore the sample would still be correctly classified.
This new value $\tilde{\xi}_j = 0$ yields a strictly smaller objective since

$$(\tilde{\xi}_j)^2 = 0 < (\xi_j^*)^2,$$

which contradicts the optimality of $\boldsymbol{\xi}^*$.
Hence, at the optimum, $\xi_n^* \geq 0$ automatically, so removing the constraints does not change the solution.

After removing the non-negativity constraints for the slack variables, we obtain the relaxed optimization problem used in the rest of the exercise:

$$\begin{aligned} \min_{\mathbf{w},b,\boldsymbol{\xi}} \quad & \frac{1}{2}\mathbf{w}^\top\mathbf{w} + \frac{C}{2}\sum_{n=1}^{N}\xi_n^2 \\ \text{subject to} \quad & t_n(\mathbf{w}^\top\mathbf{x}_n + b) \geq 1 - \xi_n, \qquad n = 1, \ldots, N. \end{aligned}$$

(b) Write the Lagrangian.

Solution:

The Lagrangian is

$$\mathcal{L}(\mathbf{w}, b, \boldsymbol{\xi}, \boldsymbol{\alpha}) = \frac{1}{2}\mathbf{w}^\top\mathbf{w} + \frac{C}{2}\sum_{n=1}^{N}\xi_n^2 - \sum_{n=1}^{N}\alpha_n\left(t_n(\mathbf{w}^\top\mathbf{x}_n + b) - 1 + \xi_n\right),$$

where $\alpha_n \geq 0$ are the Lagrange multipliers for the inequality constraints.

(c) List the KKT conditions. How many scalar conditions are there?

Solution:

The KKT conditions are:

- Primal feasibility: $t_n(\mathbf{w}^\top\mathbf{x}_n + b) - 1 + \xi_n \geq 0$ for all n.
- Dual feasibility: $\alpha_n \geq 0$ for all n.
- Complementary slackness: $\alpha_n\left(t_n(\mathbf{w}^\top\mathbf{x}_n + b) - 1 + \xi_n\right) = 0$ for all n.

There are $3N$ scalar KKT conditions in total.

(d) Derive the stationary conditions for the Lagrangian.

Solution:

Differentiating with respect to $\mathbf{w}$ gives

$$\frac{\partial\mathcal{L}}{\partial\mathbf{w}} = \mathbf{w} - \sum_{n=1}^{N}\alpha_n t_n \mathbf{x}_n.$$

Setting this to zero gives

$$\mathbf{w} = \sum_{n=1}^{N} \alpha_n t_n \mathbf{x}_n.$$

Differentiating with respect to b gives

$$\frac{\partial \mathcal{L}}{\partial b} = -\sum_{n=1}^{N} \alpha_n t_n.$$

Setting this to zero gives

$$\sum_{n=1}^{N} \alpha_n t_n = 0.$$

Finally, differentiating with respect to ξ_n gives

$$\frac{\partial \mathcal{L}}{\partial \xi_n} = C\xi_n - \alpha_n.$$

Setting this to zero gives

$$\xi_n = \frac{\alpha_n}{C}, \qquad n = 1, \ldots, N.$$

(e) Derive the dual Lagrangian and state the dual optimization problem.

Solution:

The dual Lagrangian is defined as

$$\tilde{\mathcal{L}}(\boldsymbol{\alpha}) = \inf_{\mathbf{w}, b, \boldsymbol{\xi}} \mathcal{L}(\mathbf{w}, b, \boldsymbol{\xi}, \boldsymbol{\alpha}).$$

Substituting the stationary conditions into the Lagrangian gives

$$\frac{1}{2}\mathbf{w}^\top \mathbf{w} = \frac{1}{2}\sum_{n=1}^{N}\sum_{j=1}^{N} \alpha_n \alpha_j t_n t_j \mathbf{x}_n^\top \mathbf{x}_j,$$

$$\frac{C}{2}\sum_{n=1}^{N} \xi_n^2 = \frac{1}{2C}\sum_{n=1}^{N} \alpha_n^2,$$

and

$$\begin{aligned}
&-\sum_{n=1}^{N} \alpha_n \Big(t_n(\mathbf{w}^\top \mathbf{x}_n + b) - 1 + \xi_n \Big) \\
&\quad = -\sum_{n=1}^{N} \alpha_n t_n \mathbf{w}^\top \mathbf{x}_n - b\sum_{n=1}^{N} \alpha_n t_n + \sum_{n=1}^{N} \alpha_n - \sum_{n=1}^{N} \alpha_n \xi_n \\
&\quad = -\mathbf{w}^\top \mathbf{w} + \sum_{n=1}^{N} \alpha_n - \frac{1}{C}\sum_{n=1}^{N} \alpha_n^2.
\end{aligned}$$

Putting the terms together, we obtain

$$\tilde{\mathcal{L}}(\boldsymbol{\alpha}) = \sum_{n=1}^{N} \alpha_n - \frac{1}{2}\sum_{n=1}^{N}\sum_{j=1}^{N} \alpha_n \alpha_j t_n t_j \mathbf{x}_n^\top \mathbf{x}_j - \frac{1}{2C}\sum_{n=1}^{N} \alpha_n^2.$$

Thus, the dual optimization problem is

$$\max_{\boldsymbol{\alpha}} \left[\sum_{n=1}^{N} \alpha_n - \frac{1}{2} \sum_{n=1}^{N} \sum_{j=1}^{N} \alpha_n \alpha_j t_n t_j \mathbf{x}_n^\top \mathbf{x}_j - \frac{1}{2C} \sum_{n=1}^{N} \alpha_n^2 \right]$$

subject to

$$\alpha_n \geq 0, \qquad n = 1, \ldots, N, \qquad \sum_{n=1}^{N} \alpha_n t_n = 0.$$

(f) We obtain an optimal solution $\boldsymbol{\alpha}^*$ for the dual problem. Compute the optimal primal variables $\mathbf{w}^*$, b^*, and $\boldsymbol{\xi}^*$ from that value.

Solution:

The optimal weight vector is recovered from

$$\mathbf{w}^* = \sum_{n=1}^{N} \alpha_n^* t_n \mathbf{x}_n.$$

The optimal slack variables are

$$\xi_n^* = \frac{\alpha_n^*}{C}, \qquad n = 1, \ldots, N.$$

The bias can be computed using any training point $(\mathbf{x}_i, t_i)$ with $\alpha_i^* > 0$ whose constraint is active:

$$t_i \left((\mathbf{w}^*)^\top \mathbf{x}_i + b^* \right) = 1 - \xi_i^*.$$

Solving for b^* gives

$$b^* = \frac{1 - \xi_i^*}{t_i} - (\mathbf{w}^*)^\top \mathbf{x}_i.$$

Since $t_i \in \{-1, +1\}$, this can also be written as

$$b^* = t_i (1 - \xi_i^*) - (\mathbf{w}^*)^\top \mathbf{x}_i.$$

In practice, one can average this value over several support vectors for greater numerical stability.

(g) How can we use the optimal parameters $(\mathbf{w}^*, b^*)$ to classify a test datapoint $\mathbf{x}$?

Solution:

We classify a test datapoint $\mathbf{x}$ using

$$\hat{t} = \text{sign} \left((\mathbf{w}^*)^\top \mathbf{x} + b^* \right).$$

Exercise 3: Gaussian Processes

Exam question

(a) What is a Gaussian Process? Is it a supervised or an unsupervised method?

Solution:

A GP is a specification of a probability distribution over functions, where the joint distribution of any finite set of function values is Gaussian. It is typically a **supervised** learning method used for regression and classification.

(b) In a GP, we treat the function value $f(x)$ at a specific position as a random variable. If we have two positions x and x', how is the correlation between the random variables $f(x)$ and $f(x')$ defined in a Gaussian Process?

Solution:

The correlation (or covariance) is defined by the **kernel function** (or covariance function) $k(x, x')$. This function specifies how similar the outputs $f(x)$ and $f(x')$ should be based on the relationship between inputs x and x'.

(c) Consider the standard Gaussian Radial Basis Function (RBF) or Squared Exponential kernel: $k(x, x') = \sigma^2 \exp\left(-\frac{\|x-x'\|^2}{2\ell^2}\right)$.

- What does this kernel imply about the function properties?
- Is there a parameter that controls how "smooth" or "wiggly" the sampled functions are? How would you change it to increase smoothness/correlation?

Solution:

- It implies **smoothness** and stationarity.
- The **length-scale** parameter ℓ. Increasing ℓ increases the range of correlation, making the function **smoother** (less wiggly). Decreasing ℓ makes it wiggle more rapidly.

(d) In other ML methods like Support Vector Machines (SVMs), we also use kernels (the "kernel trick"). Why do we do this?

Solution:

The kernel trick allows us to operate in a high-dimensional (or even infinite-dimensional) feature space without explicitly computing the coordinates of the data in that space. Instead of working with handcrafted features $\phi(x)$, which implicitly define a kernel $k(x, x') = \phi(x)^T \phi(x')$, we define the kernel directly, which implicitly allows us to learn non-linear boundaries efficiently.